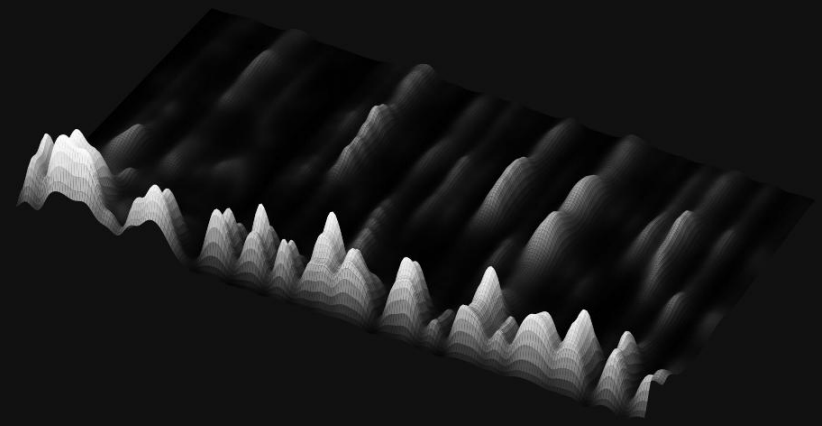
Detects AI-cloned and synthetic voices from a few seconds of audio.

1.40%
equal error rate

71,237
test clips

Voice-Shield

Real voice or AI clone? Know in a second.



Graph-attention network · runs in the browser · audio never leaves your device

Anyone's voice **can be cloned**

Seconds of audio

Free tools clone a voice from a short recording.

01

Convincing calls

Fraudsters use cloned voices to pose as a boss, a relative or a customer.

02

Ears aren't enough

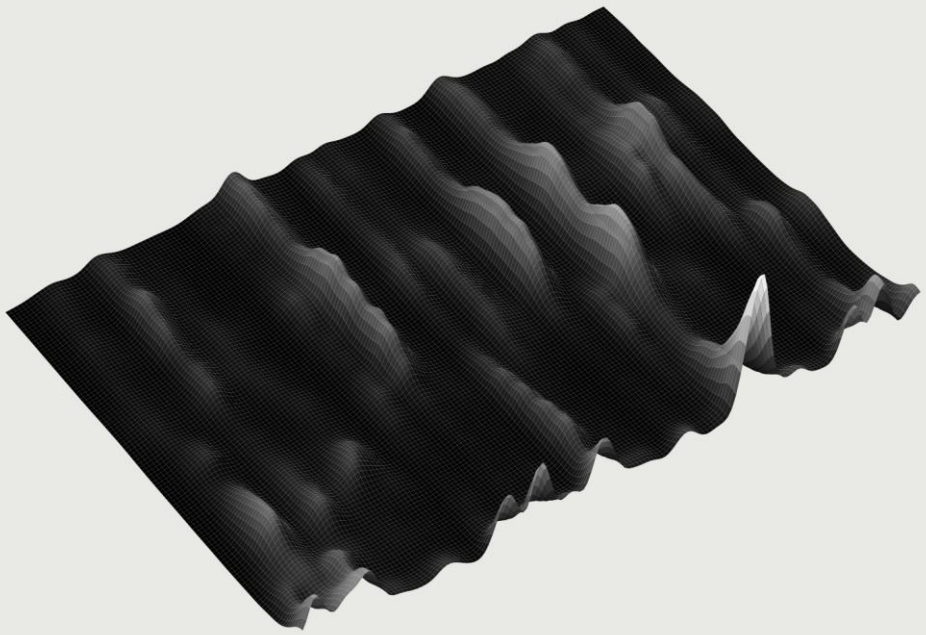
Good synthetic speech is hard to spot by listening alone.

03

Screen automatically

Call centres, banks and platforms need a fast, automatic check.

04



Real voice or AI clone?

VoiceShield listens to a few seconds of speech and flags synthetic or cloned voices, the tiny vocoder artefacts that ears miss. It runs in the browser, so recordings never leave the device.

Try the live demo →

Record · upload · test clips

Catches AI voices **it has never heard**

1.40%

equal error rate
lower is better

83.9%

AI clips caught
0.1% of real voices flagged

0.042

min t-DCF
official ASVspoof metric

1.1 s

per clip in a browser
no server, no upload

How it was measured ASVspoof 2019 LA evaluation set: 7,355 real and 63,882 AI clips from 13 attack systems, 11 of them never seen in training. The pass/fail threshold is fixed on the separate development set.

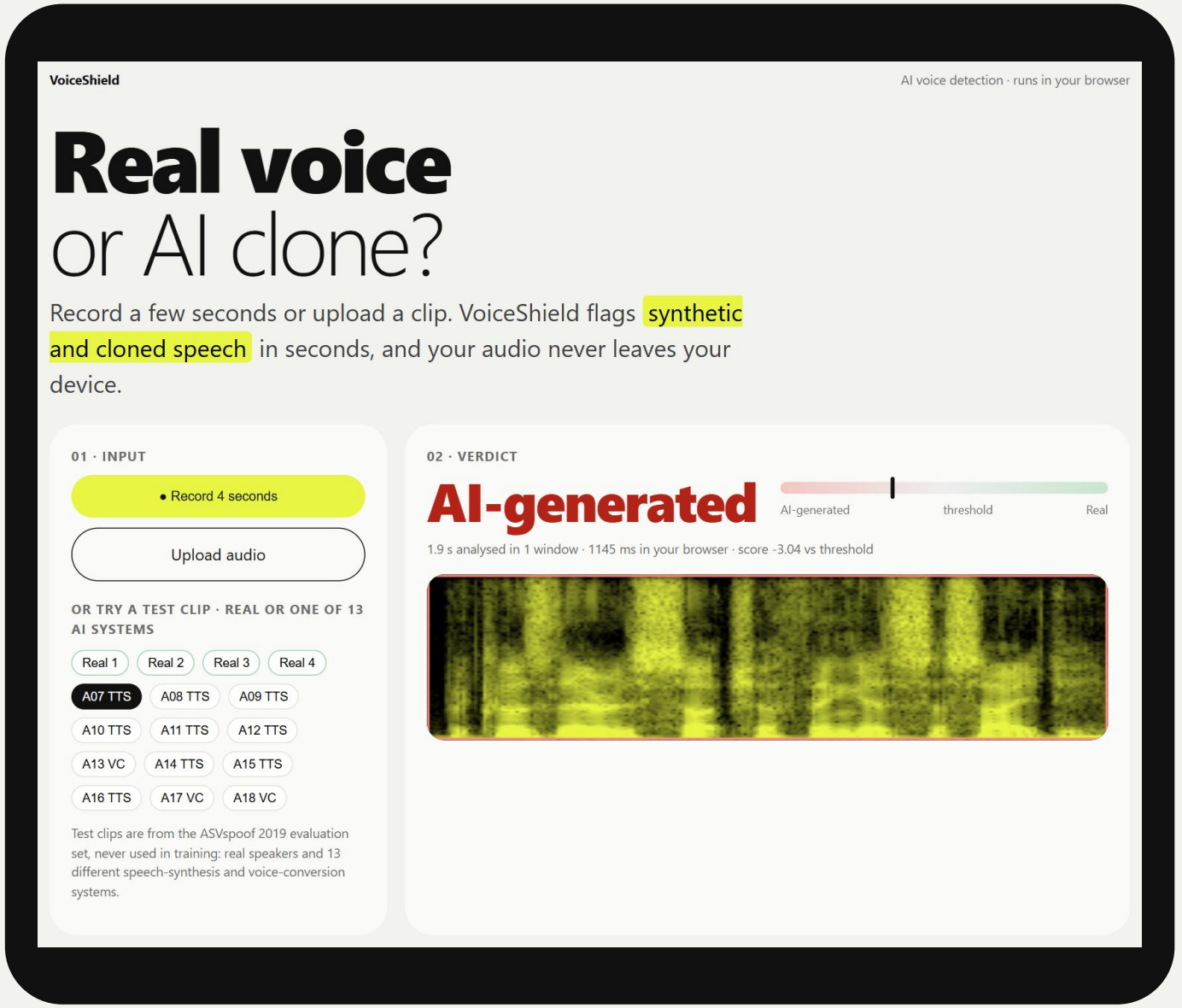
Product overview

01 **Record or upload**
4-second mic recording, or any audio file

02 **Instant verdict**
Real voice vs AI-generated, with a confidence needle

03 **See the evidence**
Spectrogram with each analysed window marked

04 **Private by design**
The model runs in the browser; nothing is uploaded



Graph attention **on raw audio**

01 **Raw waveform in**
No hand-made features: 4 s of 16 kHz audio.

02 **Learnable band-pass filters**
70 sinc filters split the sound into frequency bands.

03 **Two graphs**
One graph over frequency bands, one over time frames, each with attention.

04 **Fused by a master node**
A heterogeneous graph links both views and scores real vs AI.

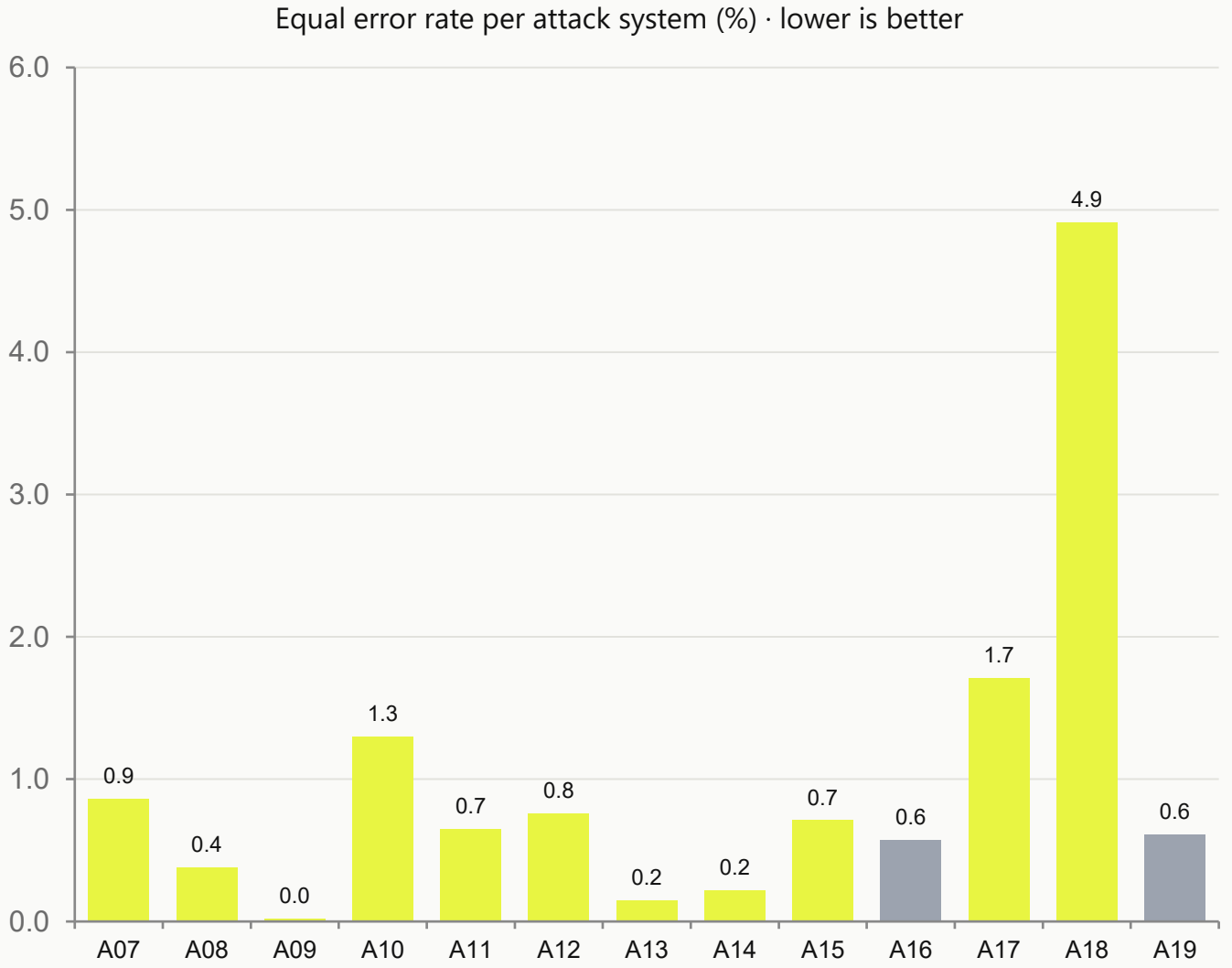
AASIST learns from raw audio which frequency bands and moments give a synthetic voice away, then lets the two graphs vote through a shared master node. 297,866 parameters: small enough to run in a web browser.

AASIST · ICASSP 2022 · built from scratch

Every attack **type, tested**

Error rate per attack system on the evaluation set. Hardest: A18 at 4.9%. Systems marked grey reuse an algorithm from the training set; the rest were never seen.

Yellow = never seen in training



Benchmarked **against the field**

Model	Parameters	EER ↓	min t-DCF ↓	Source
LFCC + LCNN (mine)	**158,146**	**5.75%**	**0.1665**	**this work · 30 epochs**
RawNet2 (mine)	**17,621,410**	**6.04%**	**0.1421**	**this work · 30 epochs**
AASIST-L (mine)	**85,306**	**1.61%**	**0.0491**	**this work · 30 epochs**
AASIST (mine)	**297,866**	**1.40%**	**0.0424**	**this work · 38 epochs**
AASIST (official model)	297k	0.83%	0.0275	clovaai/aasist (100 epochs)
AASIST-L (official model)	85k	0.99%	0.0309	clovaai/aasist (100 epochs)

My models: trained on Kaggle T4 GPUs for fewer epochs than the official recipe (12-hour session limit). Official numbers: the authors' released pretrained models, from github.com/clovaai/aasist.

A vision backed **by clear steps**

01

Send sample calls
A few minutes of your real recordings (and any known fakes).

02

Calibrated pilot
I tune the threshold on your audio and report catch rate vs false alarms.

03

Deploy
Browser, API or on-premise service: your audio stays with you.

Screen your calls **for AI voices**

Scan for the live demo

Live demo
priyanshu-singh-git.github.io/voiceshield/

Code, benchmark and limits
github.com/Priyanshu-Singh-git/voiceshield

Send me sample calls · I'll run a pilot

Priyanshu Singh · Freelance AI / ML Engineer · Data: ASVspoof 2019 (ODC-By)